

Math 118 – Fourier Analysis

Homework 2 — my solutions

Fall 2026 ■ Keshav Ramamurthy

HOMEWORK 2
due Wed, Sep. 16, 11:59 PM
(Upload your solutions to Gradescope)

1. (3 points) Let A be an $m \times n$ matrix. Prove that

$$\|A\|_{\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|.$$

2. (2 points) Suppose $A : V \rightarrow W$ and $B : U \rightarrow V$ are bounded operators. Show that $(AB)^* = B^*A^*$. If A is invertible, show that A^* is invertible and $(A^*)^{-1} = (A^{-1})^*$.
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3. (3 points) (#12 p. 35. Stop at x^2 , i.e., drop x^3 from the list.) By using the Gram-Schmidt orthogonalization procedure, find an orthonormal basis for the subspace of $L^2[0, 1]$ spanned by $1, x, x^2$.
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4. (3 points) (#13 p. 36. Stop at x^2 and replace $\cos x$ by $\cos(\pi x)$.) Find the $L^2[0, 1]$ projection of the function $\cos(\pi x)$ onto the space spanned by $1, x, x^2$.
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5. (3 points) (#15 p. 36.) Project the function $f(x) = x$ onto the space spanned by $\phi(x), \psi(x), \psi(2x), \psi(2x - 1) \in L^2[0, 1]$, where

$$\phi(x) = \begin{cases} 1, & 0 \leq x < 1, \\ 0, & \text{otherwise,} \end{cases} \quad \psi(x) = \begin{cases} 1, & 0 \leq x < 1/2, \\ -1, & 1/2 \leq x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

6. (2 points) (#19 p. 36.) Suppose $K(x, y)$ is a continuous function that vanishes outside a bounded set in $\mathbb{R}^2$. Define $T : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ by

$$Tf(x) = \int_{-\infty}^{\infty} K(x, y)f(y) dy.$$

Show that $T^*g(x) = \int_{-\infty}^{\infty} \overline{K(y, x)}g(y) dy$. Note the parallel with the adjoint of a matrix ($A_{ij}^* = \overline{A_{ji}}$).

7. (2 points) (#20 p. 36.) Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Show that $\text{Ker}(A^*) = (\text{Range}(A))^\perp$. Note: Ker stands for Kernel; $\text{Ker}(A^*)$ is the set of all vectors in W which are sent to zero by A^* .

8. (2 points) (#21 p. 37. Assume V is finite dimensional—the theorem is not generally true otherwise.) Prove the following theorem (Fredholm’s alternative): Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Let b be any element in W . Then either

- $Ax = b$ has a solution $x \in V$; or
 - There is a vector $w \in W$ with $A^*w = 0$ and $\langle b, w \rangle_W \neq 0$.
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Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggress and Narcowich.

Solutions

Problem 1

(3 points) Let A be an $m \times n$ matrix. Prove that

$$\|A\|_{\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|.$$

Solution. To show that this satisfies the properties of the norm, we go in order.

$\|A\|_{\infty} \geq 0$ and $\|A\|_{\infty} = 0$ implies $A = 0$ matrix.

This follows because the maximum sum of the absolute values in each row can only be 0 if every element of every row is 0.

It is also clear that $\|A\|_{\infty} \geq 0$ because it is a sum of absolute values.

We wish to show that $\|cA\|_{\infty} = |c|\|A\|_{\infty}$.

$$\begin{aligned} \|cA\|_{\infty} &= \max_{1 \leq i \leq m} \sum_{j=1}^n |cA_{ij}| \\ &= \max_{1 \leq i \leq m} \sum_{j=1}^n |c| |A_{ij}| \\ &= |c| \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}| \\ &= |c| \|A\|_{\infty}. \end{aligned}$$

Lastly, we wish to show the triangle inequality. Using the usual triangle inequality,

$$\begin{aligned} \|A + B\|_{\infty} &= \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij} + B_{ij}| \\ &\leq \max_{1 \leq i \leq m} \sum_{j=1}^n (|A_{ij}| + |B_{ij}|) \\ &\leq \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}| + \max_{1 \leq i \leq m} \sum_{j=1}^n |B_{ij}| \\ &= \|A\|_{\infty} + \|B\|_{\infty}. \end{aligned}$$

So all the conditions hold and this is a valid norm. □

Problem 2

(2 points) Suppose $A : V \rightarrow W$ and $B : U \rightarrow V$ are bounded operators. Show that $(AB)^* = B^*A^*$. If A is invertible, show that A^* is invertible and $(A^*)^{-1} = (A^{-1})^*$.

Solution. Starting from the adjoint formula, $\langle Tv, w \rangle = \langle v, T^*w \rangle$.

We have $\langle ABv, w \rangle = \langle v, (AB)^*w \rangle$.

We can separate A and B using the adjoint formula:

$$\langle ABv, w \rangle = \langle Bv, A^*w \rangle = \langle v, B^*A^*w \rangle.$$

Thus, $(AB)^* = B^*A^*$.

If A is invertible, then $A^{-1}A = I$. Applying the adjoint to both sides gives $A^*(A^{-1})^* = I$.

Similarly, from $AA^{-1} = I$, we get $(A^{-1})^*A^* = I$.

Thus $(A^{-1})^*$ is both a left and right inverse of A^* , so $(A^*)^{-1} = (A^{-1})^*$. $\square$

Problem 3

(3 points) (#12 p. 35. Stop at x^2 , i.e., drop x^3 from the list.) By using the Gram–Schmidt orthogonalization procedure, find an orthonormal basis for the subspace of $L^2[0, 1]$ spanned by $1, x, x^2$.

Solution. We start with $v_1 = 1$. Its norm is $\sqrt{\int_0^1 1 \, dx} = 1$, so our first basis vector is 1.

Then, we consider x . Using our inner product, $\int_0^1 1x \, dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$, so we subtract off $\frac{1}{2}$ to get $x - \frac{1}{2}$.

Now we normalize $x - \frac{1}{2}$:

$$\sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 \, dx} = \frac{1}{2\sqrt{3}}.$$

Thus the normalized vector is $2\sqrt{3}\left(x - \frac{1}{2}\right)$.

Now consider x^2 . Starting with 1, we find $\int_0^1 1x^2 \, dx = \frac{x^3}{3} \Big|_0^1 = \frac{1}{3}$, so we get $x^2 - \frac{1}{3}$.

Using the second basis vector,

$$\int_0^1 2\sqrt{3}\left(x - \frac{1}{2}\right)\left(x^2 - \frac{1}{3}\right) \, dx = \frac{1}{2\sqrt{3}}.$$

Therefore,

$$\left(x^2 - \frac{1}{3}\right) - \frac{1}{2\sqrt{3}}2\sqrt{3}\left(x - \frac{1}{2}\right) = x^2 - x + \frac{1}{6}.$$

Finally, $\int_0^1 (x^2 - x + \frac{1}{6})^2 dx = \frac{1}{180}$, so the normalized vector is $6\sqrt{5} (x^2 - x + \frac{1}{6})$.

Thus an orthonormal basis is 1 , $2\sqrt{3} (x - \frac{1}{2})$, and $6\sqrt{5} (x^2 - x + \frac{1}{6})$. $\square$

Problem 4

(3 points) (#13 p. 36. Stop at x^2 and replace $\cos x$ by $\cos(\pi x)$.) Find the $L^2[0, 1]$ projection of the function $\cos(\pi x)$ onto the space spanned by $1, x, x^2$.

Solution. We want to find the $L^2[0, 1]$ projection of $\cos(\pi x)$ onto the space spanned by $1, x, x^2$.

Since we already found an orthonormal basis for this space, we can use the projection formula. Let $f(x) = \cos(\pi x)$ and 1 , $2\sqrt{3} (x - \frac{1}{2})$, and $6\sqrt{5} (x^2 - x + \frac{1}{6})$ be our orthonormal basis.

First,

$$\int_0^1 \cos(\pi x) dx = \left. \frac{\sin(\pi x)}{\pi} \right|_0^1 = 0.$$

Next,

$$\begin{aligned} \int_0^1 2\sqrt{3} \left(x - \frac{1}{2} \right) \cos(\pi x) dx &= 2\sqrt{3} \left(\int_0^1 x \cos(\pi x) dx - \frac{1}{2} \int_0^1 \cos(\pi x) dx \right) \\ &= 2\sqrt{3} \left(-\frac{2}{\pi^2} - 0 \right) \\ &= -\frac{4\sqrt{3}}{\pi^2}. \end{aligned}$$

Finally,

$$\begin{aligned} \int_0^1 6\sqrt{5} \left(x^2 - x + \frac{1}{6} \right) \cos(\pi x) dx &= 6\sqrt{5} \left(\int_0^1 x^2 \cos(\pi x) dx - \int_0^1 x \cos(\pi x) dx + \frac{1}{6} \int_0^1 \cos(\pi x) dx \right) \\ &= 6\sqrt{5} \left(-\frac{2}{\pi^2} + \frac{2}{\pi^2} + 0 \right) \\ &= 0. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{proj}(f) &= 0(1) - \frac{4\sqrt{3}}{\pi^2} \left[2\sqrt{3} \left(x - \frac{1}{2} \right) \right] \\ &= -\frac{24}{\pi^2} \left(x - \frac{1}{2} \right). \end{aligned}$$

$\square$

Problem 5

(3 points) (#15 p. 36.) Project the function $f(x) = x$ onto the space spanned by

$$\varphi(x), \psi(x), \psi(2x), \psi(2x-1) \in L^2[0, 1],$$

where

$$\varphi(x) = \begin{cases} 1, & 0 \leq x < 1, \\ 0, & \text{otherwise,} \end{cases} \quad \psi(x) = \begin{cases} 1, & 0 \leq x < 1/2, \\ -1, & 1/2 \leq x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Solution. We claim that these functions are all orthogonal. We can see this by computing their pairwise inner products.

First,

$$\langle \varphi, \psi \rangle = \int_0^{1/2} (1)(1) dx + \int_{1/2}^1 (1)(-1) dx = \frac{1}{2} - \frac{1}{2} = 0.$$

Next, $\psi(2x)$ is piecewise 1, -1, 0 when $0 \leq x < \frac{1}{4}$, $\frac{1}{4} \leq x < \frac{1}{2}$, and $\frac{1}{2} \leq x \leq 1$, respectively. Therefore,

$$\langle \varphi, \psi(2x) \rangle = \int_0^{1/4} (1)(1) dx + \int_{1/4}^{1/2} (1)(-1) dx + \int_{1/2}^1 (1)(0) dx = \frac{1}{4} - \frac{1}{4} + 0 = 0.$$

Similarly, $\psi(2x-1)$ is piecewise 0, 1, -1 when $0 \leq x < \frac{1}{2}$, $\frac{1}{2} \leq x < \frac{3}{4}$, and $\frac{3}{4} \leq x < 1$, respectively. Thus,

$$\langle \varphi, \psi(2x-1) \rangle = \int_0^{1/2} (1)(0) dx + \int_{1/2}^{3/4} (1)(1) dx + \int_{3/4}^1 (1)(-1) dx = 0 + \frac{1}{4} - \frac{1}{4} = 0.$$

Now consider ψ and $\psi(2x)$. We get

$$\langle \psi, \psi(2x) \rangle = \int_0^{1/4} (1)(1) dx + \int_{1/4}^{1/2} (1)(-1) dx + \int_{1/2}^1 (-1)(0) dx = \frac{1}{4} - \frac{1}{4} + 0 = 0.$$

Next,

$$\langle \psi, \psi(2x-1) \rangle = \int_0^{1/2} (1)(0) dx + \int_{1/2}^{3/4} (-1)(1) dx + \int_{3/4}^1 (-1)(-1) dx = 0 - \frac{1}{4} + \frac{1}{4} = 0.$$

Finally,

$$\begin{aligned}
\langle \psi(2x), \psi(2x-1) \rangle &= \int_0^{1/4} (1)(0) dx + \int_{1/4}^{1/2} (-1)(0) dx \\
&\quad + \int_{1/2}^{3/4} (0)(1) dx + \int_{3/4}^1 (0)(-1) dx \\
&= 0.
\end{aligned}$$

Thus they are all orthogonal, and we can add the individual projections of $f(x) = x$ onto each function.

$$\text{proj}_\varphi(f) = \frac{\int_0^1 x dx}{\int_0^1 1 dx} \varphi = \frac{1}{2} \varphi$$

$$\text{proj}_\psi(f) = \frac{\int_0^{1/2} x dx - \int_{1/2}^1 x dx}{\int_0^{1/2} 1 dx + \int_{1/2}^1 (-1)^2 dx} \psi = -\frac{1}{4} \psi$$

$$\text{proj}_{\psi(2x)}(f) = \frac{\int_0^{1/4} x dx - \int_{1/4}^{1/2} x dx}{\int_0^{1/4} 1 dx + \int_{1/4}^{1/2} (-1)^2 dx} \psi(2x) = -\frac{1}{8} \psi(2x)$$

$$\text{proj}_{\psi(2x-1)}(f) = \frac{\int_{1/2}^{3/4} x dx - \int_{3/4}^1 x dx}{\int_{1/2}^{3/4} 1 dx + \int_{3/4}^1 (-1)^2 dx} \psi(2x-1) = -\frac{1}{8} \psi(2x-1)$$

$$\text{proj}(f) = \frac{1}{2} \varphi - \frac{1}{4} \psi - \frac{1}{8} \psi(2x) - \frac{1}{8} \psi(2x-1).$$

□

Problem 6

(2 points) (#19 p. 36.) Suppose $K(x, y)$ is a continuous function that vanishes outside a bounded set in $\mathbb{R}^2$. Define $T : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ by

$$Tf(x) = \int_{-\infty}^{\infty} K(x, y) f(y) dy.$$

Show that $T^*g(x) = \int_{-\infty}^{\infty} K(y, x) g(y) dy$. Note the parallel with the adjoint of a matrix ($A_{ij}^* = A_{ji}$).

Solution. Starting from the adjoint formula, $\langle Tf, g \rangle = \langle f, T^*g \rangle$.

$$\begin{aligned}
\langle Tf, g \rangle &= \int_{-\infty}^{\infty} Tf(x)g(x) dx \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} K(x, y)f(y) dy \right) g(x) dx \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K(x, y)f(y)g(x) dy dx \\
&= \int_{-\infty}^{\infty} f(y) \left(\int_{-\infty}^{\infty} K(x, y)g(x) dx \right) dy.
\end{aligned}$$

Since x is a dummy variable and y is the input variable, we rename the variables to get $T^*g(x) = \int_{-\infty}^{\infty} K(y, x)g(y) dy$. $\square$

Problem 7

(2 points) (#20 p. 36.) Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Show that $\ker(A^*) = (\text{Range}(A))^\perp$. Note: Ker stands for Kernel; $\ker(A^*)$ is the set of all vectors in W which are sent to zero by A^* .

Solution. Let $y \in \ker(A^*)$. Then $A^*y = 0$, so for any x using the adjoint formula,

$$\langle Ax, y \rangle = \langle x, A^*y \rangle = \langle x, 0 \rangle = 0.$$

Thus y is orthogonal to every element in $\text{Range}(A)$, so $y \in \text{Range}(A)^\perp$.

Therefore, $\ker(A^*) \subseteq \text{Range}(A)^\perp$.

Now, let $y \in \text{Range}(A)^\perp$. Then $\langle Ax, y \rangle = 0$ for every $x \in V$. Using the adjoint, $\langle x, A^*y \rangle = 0$ for every $x \in V$, which implies $A^*y = 0$.

Thus $y \in \ker(A^*)$, so $\text{Range}(A)^\perp \subseteq \ker(A^*)$.

Therefore, $\ker(A^*) = \text{Range}(A)^\perp$. $\square$

Problem 8

(2 points) (#21 p. 37. Assume V is finite dimensional — the theorem is not generally true otherwise.) Prove the following theorem (Fredholm's alternative): Suppose $A : V \rightarrow W$ is a linear map between two inner product spaces. Let b be any element in W . Then either

- $Ax = b$ has a solution $x \in V$; or
- there is a vector $w \in W$ with $A^*w = 0$ and $\langle b, w \rangle_W \neq 0$.

Solution. We'll do casework on whether b is in $\text{Range}(A)$.

If $b \in \text{Range}(A)$, then by definition there is some $x \in V$ such that $Ax = b$, so the first case holds.

Otherwise, $b \notin \text{Range}(A)$. Since V is finite dimensional, $\text{Range}(A)$ is finite dimensional, so $W = \text{Range}(A) \oplus \text{Range}(A)^\perp$.

Therefore, since $b \notin \text{Range}(A)$, it has a nonzero orthogonal part. Thus there is some $w \in \text{Range}(A)^\perp$ with $w \neq 0$ such that $b = s + w$ for some $s \in \text{Range}(A)$. Therefore,

$$\langle b, w \rangle_W = \langle s + w, w \rangle_W = \langle s, w \rangle_W + \langle w, w \rangle_W = 0 + \|w\|^2 > 0.$$

Since $w \in \text{Range}(A)^\perp$, we have $\langle Ax, w \rangle_W = 0$ for every $x \in V$. Using the adjoint formula, $\langle x, A^*w \rangle_V = 0$ for every $x \in V$, so $A^*w = 0$.

Therefore, if $b \in \text{Range}(A)$, then $Ax = b$ has a solution. Otherwise, there is a vector w satisfying $A^*w = 0$ and $\langle b, w \rangle_W \neq 0$.

So either $Ax = b$ has a solution, or there is a vector w with $A^*w = 0$ and $\langle b, w \rangle_W \neq 0$. □

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